

Reviewer #1

Response to Review-1 Feedback to all Q

Q-1

One of the key improvements involved removing the manually set weights for the ensemble approach. Initially, the weight values were assigned empirically without any supporting rationale. In response to the feedback, we have now implemented a dynamic method for setting the weights. This approach allows the model to determine the appropriate weights automatically based on the performance of each individual model in the ensemble. This improvement not only enhances the adaptability of the model but also ensures a more data-driven and optimized weight assignment.

The review pointed out the absence of detailed descriptions of the individual models used in the ensemble. In response, we have added comprehensive descriptions for each model, including BanglaBERT, LSTM, BiLSTM, and GRU. Additionally, we now provide an overview of the relevant parameters for each model, offering clarity on their specific configurations. These updates help readers better understand the models' architectures and their roles within the ensemble approach.

Q-2

Table Caption: In response to the reviewer's suggestion regarding table captions, we have corrected the positioning of the captions for Table 4 and Table 5. Initially, the captions were placed at the bottom of the tables, which is not in line with standard formatting conventions. Following the feedback, we have moved the captions to the top of the tables, ensuring consistency with academic presentation guidelines. This change improves the clarity and readability of the tables, making the paper more professional and easier to follow for the readers.

Q-3

In response to the reviewer's feedback on the impact of the ensemble approach, we have added a new "**Abilarion Study**" subsection. This section provides a detailed discussion of each model used in the ensemble (BanglaBERT, LSTM, BiLSTM, and GRU) and examines the effect of removing each model from the ensemble one by one. By evaluating the performance degradation when individual models are excluded, we highlight the unique contribution of each model and demonstrate the overall benefit of combining them in the ensemble.

Additionally, we have included an "**Ensemble Weight Analysis**" section, where we explain how dynamic weights are assigned to each model based on datalength, further justifying the strength of the ensemble approach.

These enhancements clarify the impact of the ensemble method and show how it improves sentiment analysis results.

Q-4

In response to the reviewer's comment, we maintain the primary focus on the proposed ensemble approach, as no existing studies directly combine the models used in our work. While we appreciate the suggestion to include other benchmark studies, we believe the novelty of our ensemble method stands on its own. Instead of broadening the scope, we have refined the analysis of the ensemble's performance and its unique advantages in sentiment analysis for Bangla E-commerce reviews.

Reviewer #2

Review-2 not visible in the portal.

Reviewer #3

Review-3 is : However, more details on error analysis and real-world applicability (e.g., how this could be used in live systems or platforms) would strengthen the paper. A brief ablation study to show the contribution of each component would also be valuable further proving the importance of BanglaBERT. Language clarity is good, though a more concise abstract would improve readability.

Response to Review-3 Feedback

In response to the reviewer's suggestion, we have added a new subsection focusing on **error analysis**. This section provides a deeper examination of the types of errors made by the model and offers insights into how they could be addressed. By highlighting the areas where the model struggles, we aim to demonstrate a clear path for future improvements and refinement in the approach. [paper page- 11]

Additionally, we have introduced a **real-world applicability Case study** subsection to show how our model can be applied in live systems or platforms, such as E-commerce websites. This section emphasizes the practical benefits of using our ensemble approach in real-world scenarios, explaining how it can enhance customer experience through better sentiment analysis of product reviews. [paper page- 12]

Finally, we have included a brief **ablation study**, which analyzes the contribution of each component in the ensemble. This helps to highlight the importance of each individual model, particularly BanglaBERT, and demonstrates how their combined use improves overall performance. We also made minor improvements to the abstract to make it more concise, improving the clarity and readability of the paper. [paper page- 10]

Reviewer #4

Review-4: Suggestions for making things better:

1. **Actual Application:** A small case study or demonstration of how the model may be used on a Bangla e-commerce platform will make the research more useful in real life.
2. **Ablation research:** There are baseline comparisons, but an ablation research showing how each ensemble model adds to the hybrid approach would be helpful.

Response to Review-4 Feedback

In response to your suggestions, we have made the following improvements to the paper:

1. **Actual Application:** We have added a small **case study** demonstrating how the proposed model can be applied to a **Bangla e-commerce platform**. This case study showcases how the ensemble approach can be used in real-life systems to analyze product reviews, improve sentiment classification, and enhance customer experience. By providing this example, we aim to highlight the practical utility of the model in real-world settings. [paper page- 12]
2. **Ablation Research:** Following your recommendation, we have included an **ablation study** that investigates how each individual model contributes to the hybrid ensemble approach. This section demonstrates how the combination of BanglaBERT, LSTM, BiLSTM, and GRU results in improved performance and explains the unique contribution of each model in the ensemble. This analysis helps further validate the effectiveness of the ensemble method. [paper page- 10]